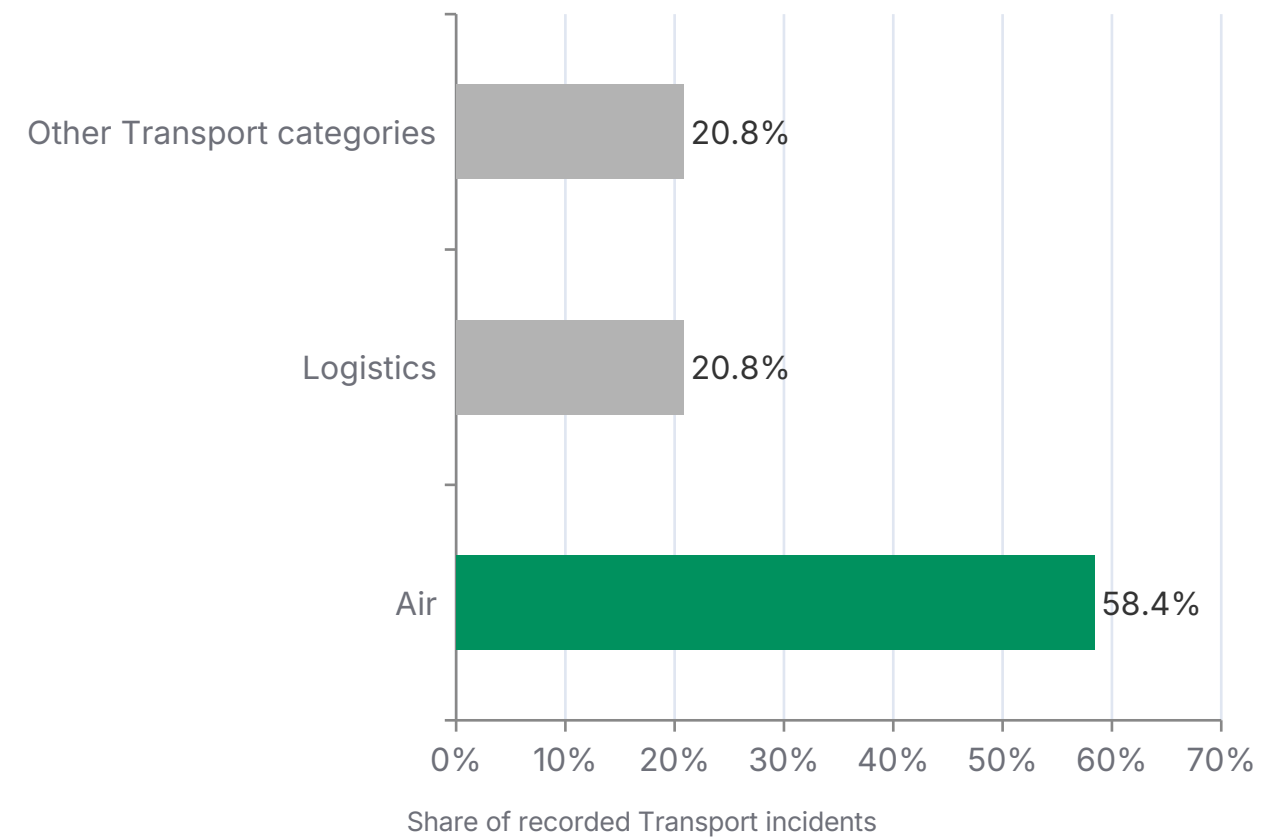


Key Threats & Top 5 Cyber Attacks

Recorded Transport subset - curated incident corpus

ENISA selected-corpus period: 01 Jul 2024 to 30 Jun 2025. Categories shown as shares of the Transport subset; data are not a complete incident census.



DDoS accounted for 87.6% of recorded Transport incidents in the same curated corpus.

Evidence-led pattern	Operational vulnerability - concentration implication
1. Shared passenger-processing disruption	Common check-in and boarding dependency - disruption can transmit across provider users and airport-community processes. [C08]
2. DDoS against rail channels	Public information and booking dependency - channel congestion and impaired customer communication. [C09]
3. Freight phishing and cargo diversion	Identity and transaction-validation weakness - cross-border fraud can propagate through freight platforms. [C10]
4. State-linked multi-vector targeting	Identity, public-service and relationship exposure - cross-modal pathway through logistics and technology dependencies. [C11]
5. Supplier access and OT/IT interfaces	Incomplete mapping and third-party access - correlated restoration difficulty across connected operations. [C06]

Financial Impact: Transport & Storage

Pathway analysis - no EU loss total

Public evidence supports transmission pathways, not a current EU sector loss estimate.

1. Operational disruption	2. Financial transmission	3. Regulatory process	4. Insurance and liquidity timing
Passenger processing, booking, cargo, freight and OT/IT dependencies can interrupt core services. [C06, C08]	Lost or delayed revenue can occur while incident response, manual operations and restoration costs rise. [C12, C13]	NIS2 staged reporting may apply where entity scope and incident thresholds are met. [C02]	Business interruption and contingent interruption depend on cover, limits, exclusions, deductibles and claim evidence. [C15]
Manual and alternative workarounds can preserve portions of service while reducing throughput. [C08]	Customer, shipper, airport-community and counterparty relationships can transmit loss. [C15]	GDPR assessment and the 72-hour controller notification expectation apply only where legal conditions are met. [C16]	EIOPA identifies silent-cover uncertainty and possible litigation when reimbursement is disputed. [C15]
Supplier outage or access compromise can slow containment and coordinated restoration. [C06]	Maersk's historical global loss categories were lost revenue, IT restoration and extraordinary operational cost. Not used as an EU benchmark. [C13]	NIS2 fine maxima are statutory ceilings, not expected outcomes. [C17]	Cash pressure depends on daily funding needs, timing of revenue and claim receipts, and committed facilities. [C15]

Decision boundary: No current public EU Transport & Storage dataset links cyber events to final loss, claims, insurance recovery and liquidity. Entity quantification requires internal volumes, contracts, recovery and cash data.

Financial Impact: Worst-Case Scenarios

Plausible stress pathways - not observed outcomes

Stress testing must target correlated disruption and cash timing rather than a single assumed cyber-loss figure.

Scenario 1 - Shared airport-service outage	Scenario 2 - Supplier or OT/IT interface compromise	Scenario 3 - Rail and maritime disruption with data exposure
Operational effect. A common passenger-processing provider is unavailable; manual or alternative processing is activated; airlines cancel or reschedule flights. [C08] Financial pathway. Revenue timing, operating expenditure, customer-management cost, contract exposure and conditional insurance recovery move together. [C12, C15] Boundary. Multi-airport effect is a dependency scenario, not a full observed account.	Operational effect. Supplier access, fragmented mapping and multi-vendor dependencies delay containment, patching and restoration. [C06] Financial pathway. Recovery cost, service degradation, customer claims, incident-reporting assessment and conditional risk transfer overlap. [C15] Boundary. This is an ENISA exposure assessment, not a prediction for a named operator.	Operational effect. Cargo, navigation, rail and ticketing functions are disrupted across borders while passenger or emergency data is exposed. [C14] Financial pathway. Concurrent business interruption, restoration, data and claims processes test funding and insurer timing. [C15, C16] Boundary. Cyber Europe 2026 supplies the pathway. The effects are simulated.

Required inputs: common-provider map - maximum tolerable outage - manual capacity - daily cash forecast - contract terms - current insurance wording.

Case Study: Brussels Airport, Belgium: September 2025 disruption of Collins Aerospace passenger-processing services

A provider outage caused cancellations while manual and alternative processing preserved many flights.

Evidence-led timeline

20 to 28 Sep 2025 - flights cancelled

Brussels Airport disclosed that the cyberattack affected Collins Aerospace, its external provider for check-in and boarding systems. [C08]

During disruption - manual and alternative processing

The airport reported alternative systems and airport-community actions that enabled the vast majority of flights to operate. [C08]

09 Oct 2025 - replacement system in use

The airport reported accelerated rollout of a new check-in and boarding system, then used by all airlines. [C08]

2LoD interpretation

The case illustrates supplier-dependency ownership, manual fallback, incident communications and recovery decision rights. It is not a control assessment of either organisation.

Observed facts vs not established

Observed	Not established by public evidence
Provider supported airport check-in and boarding systems. [C08]	Entry route, actor identity, data access or removal. [C08]
Flights were cancelled from 20 to 28 September 2025. [C08]	Incident-specific financial impact, total cancellation effect and full affected-airport list. [C08]
Replacement check-in and boarding system was accelerated and used by airlines. [C08]	Whether replacement addressed the cause of compromise or residual exposure. [C08]

Management lesson: dependency ownership, tested fallback, supplier communications and recovery decision rights are the relevant control objectives.

Mitigation Strategies & Security Framework

2LoD control synthesis - not an entity finding

Resilience requires legal-perimeter clarity, common-dependency control, identity and OT governance, tested recovery and integrated financial response.

1. Legal and accountability mapping	2. Common-dependency resilience	3. Identity and transaction integrity	4. OT and IT recovery governance	5. Financial and incident integration
Map NIS2, CER and aviation regime by legal entity, function, country and accountable owner. [C01 to C05]	Map shared passenger, freight, booking, identity, cloud, remote-support and OT dependencies. Test alternatives and exit options. [C06, C08]	Strengthen phishing resilience, privileged access, multi-factor authentication, freight-transaction validation and cargo-release controls. [C10, C11]	Maintain asset and dependency mapping. Segment operational systems. Exercise containment, manual operations and restoration with suppliers. [C06, C14]	Pre-agree cash, customer, legal, insurance and notification workflows. Test funding and conditional insurance assumptions. [C02, C12, C15, C16]
Outcome: clear scope, reporting route and management-body oversight.	Outcome: reduced uncertainty around provider outage and recovery ownership.	Outcome: reduced credential-led fraud and supplier-access exposure.	Outcome: faster isolation and recovery across connected environments.	Outcome: credible response under concurrent operational, financial and regulatory pressure.
Committee challenge: which entity has confirmed scope and which is pending legal review?	Committee challenge: which common dependency lacks a tested alternative or exit?	Committee challenge: which freight release or payment path can proceed after credential compromise?	Committee challenge: which OT/IT pathway has not been jointly restored with key suppliers?	Committee challenge: how long can recovery spend continue before revenue and claims receipts catch up?

Decision boundary: the framework does not assert that any named entity lacks these controls. Effectiveness requires entity-specific testing and evidence.